

Name: _____

Class: _____

Index: _____

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION

ANSWERS

BIOLOGY
Paper 2 Core Paper

9747/02
Wednesday 27 August 2008

2 hours

Additional materials: Answer booklet
 Additional Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A Structured Questions

Answer **all** questions.
Write your answers in the spaces provided on the question paper.
Show All working

Section B Essay Questions

Answer all questions.
Write your answers on the separate answer paper.
All working for numerical answers must be shown.

At the end of the examination,

1. fasten all your work securely together;
2. enter the number of the Section C question you have answered in the grid opposite.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	80
1 [11]	
2 [13]	
3 [16]	
4 [10]	
5 [17]	
6 [13]	
Section B	20
7 [20]	
8 [20]	
Total	100

This document consists of **14** printed pages.

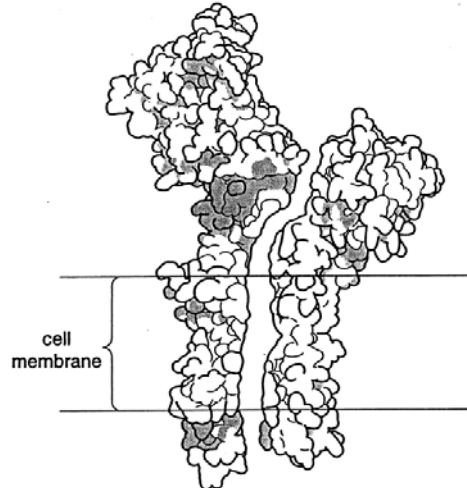
[Turn over]

Section A

ANSWERS

- 1 Fig. 1.1 shows the molecular structure of a protein pump that moves calcium ions across cell membranes.

Fig 1.1



- (a) Explain why such pumps are often necessary to move calcium ions across membranes. [3]
1. cell membrane comprises of a **phospho-lipid bilayer** which contains a **hydrophobic region** making the membrane **impermeable to hydrophilic substances**.
 2. **Calcium ions are hydrophilic** and are thus impermeable to the membrane, there is therefore a need for **Integral proteins** that span the width of the membrane, to move calcium ions across the membrane,
 3. **transport of ions against a concentration gradient**.
- (b) Outline how such a protein pump would move calcium ions across the membrane shown. [2]
1. Transport of ions is an **active process** requiring **energy i.e. ATP**
 2. Calcium ions **bind to the pumps**, causing a **conformational change** in the pump activating ATPase and moving the ion across the membrane thru the **hydrophilic channel of the pump**.

Transmission of an impulse across a synapse is triggered by a sudden influx of calcium ions into the synaptic knob. This does not occur through a protein pump.

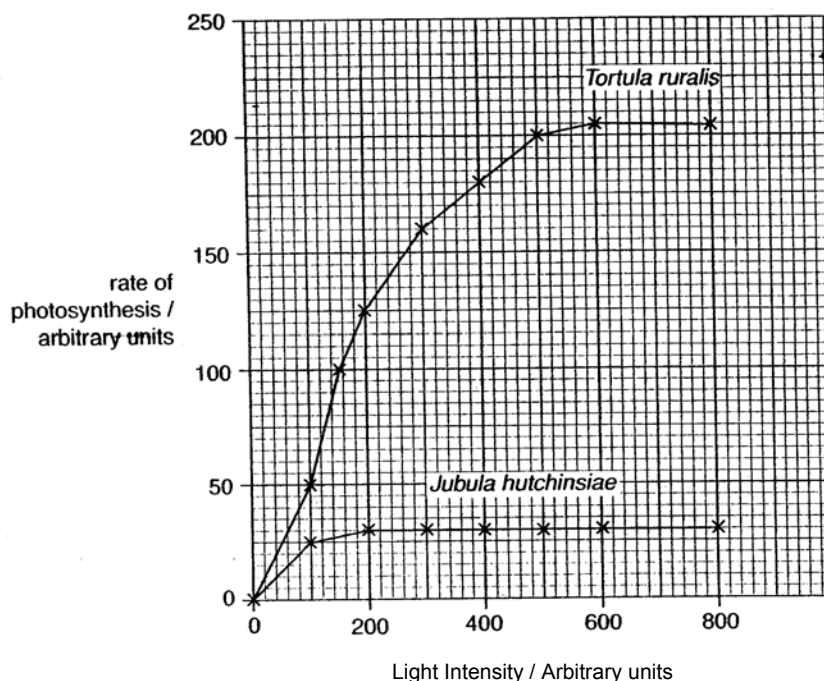
- (c) Describe how this influx of calcium ions occurs through the presynaptic membrane. [2]
1. upon the arrival of the **Action Potential, which causes a depolarization** of the membrane of the presynaptic knob, which causes the **opening of voltage gated calcium channels**
 2. calcium ions enter the presynaptic knob **moving down the concentration or diffusion gradient** into the presynaptic knob.
- (d) Explain how the entry of calcium ions into the synaptic knob leads to the impulse being transmitted across the synapse. [4]
1. calcium ions entering the presynaptic knob **cause vesicles containing transmitter substances to move to and fuse to the presynaptic membrane**.
 2. this causes an emptying **of the vesicle contents into the synaptic cleft**, neurotransmitter substances then **diffuses across** and
 3. **binds to receptors** on the **post synaptic membrane** which cause a conformational change **causing the opening of gated Na⁺ ion channels**
 4. influx of Na causes a **depolarizing of the post synaptic membrane** effecting an **action potential** in the post synaptic knob.

[Total: 11]

- 2 Manybryophytes grow in shady conditions in woodlands and forests where the lack of direct sunlight and the cool, moist conditions reduce the risk of desiccation. However, some species do survive in open, sunny habitats.

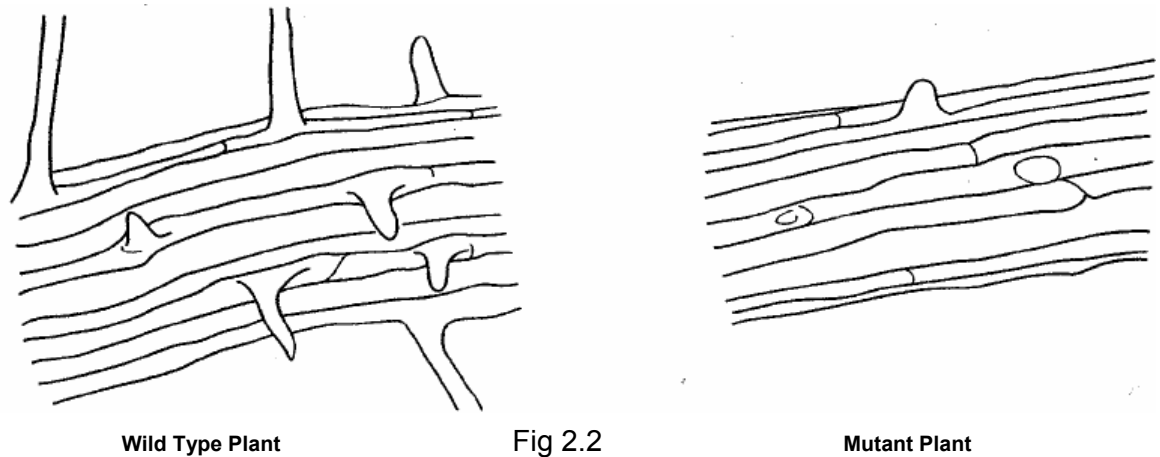
The figure below shows how the rate of photosynthesis varies with light intensity, for two bryophytes, *Jubula hutchinsiae* grows in forests, while *Tortula ruralis* grows in sunny, exposed habitats.

Fig 2.1



- (a) (i) With reference to Figure 2.1, describe how light intensity affects the rate of photosynthesis of *T. ruralis*. [2]
- As **Light intensity increases (from 0 to 600)**, **rate of photosynthesis increases (from 0 to 205 a.u.)**; **Rate of increase slows** until Maximum rate of photosynthesis reached at 205 arbitrary units from **light intensity 600** arbitrary units onwards;
 - Further increase in light intensity (from 600 to 800) **does not increase the rate of photosynthesis/** rate remained constant/plateau;
- (ii) explain the effect you have described in (2ai) [2]
- Light is an **absolute requirement** for PS, since when LI=0, rate of PS=0; Between 0-600, **light is the limiting factor**; Increase LI, increase the **light reaction/ light-dependent reaction**;
 - as more electrons are boosted to higher energy levels/ more electrons boosted from PSI and PSII; At very high light intensity of >600 a.u., light is no longer limiting, **other factors** such as CO₂ concentration or temperature is **limiting**;
- Students may break the part from 0-600 LI into 0-200 and 200-600 or thereabout to discuss about LI being a limiting factor and other factors becoming limiting.
- (iii) Suggest with reasons, why the rate of photosynthesis of *J. hutchinsiae* reaches a plateau at a much lower light intensity than that of *T. ruralis*. [2]
- J. hutchinsiae* grows in **shady low light intensity areas/** received lower amount of LI; **Adapted to the low light conditions**;
 - Lower rate of photosynthesis than *T. ruralis* since there is lower amount of light available (for light reaction to take place); **Reaches its maximum rate of PS at a lower LI (200)** so that it can make **more efficient use of light**;
 - adaptation can also come in the form of lower number of chloroplasts/ chlorophyll (as formation of chloroplasts requires light to be present in the first place);

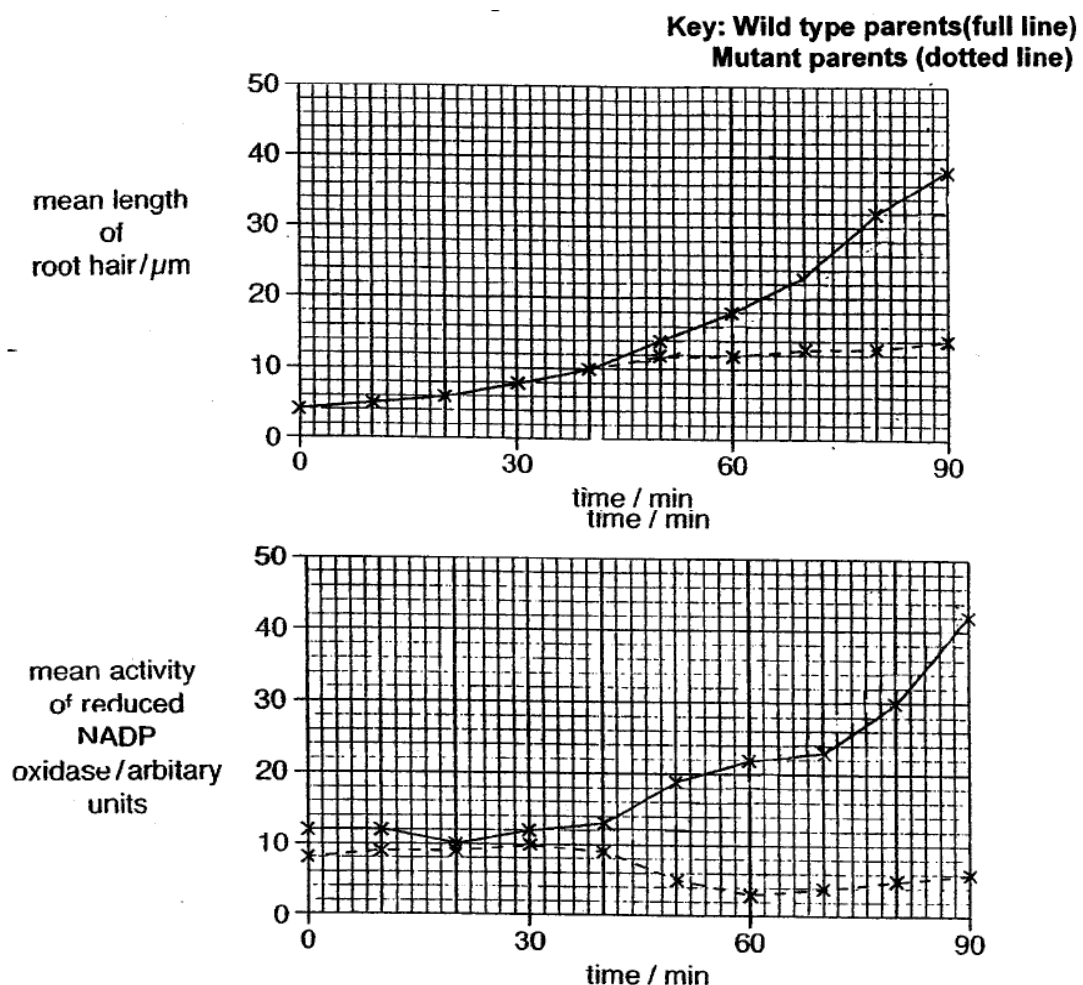
- (b) In another species of plant called *Arabidopsis thaliana* also found in the forested areas, a mutation of a gene coding for the enzyme reduced NADP oxidase results in plants with very very short root hair. The root hair of wild type and mutant plants are shown in Fig 2.2.



Reduced NADP oxidase converts reduced NADP to NADP. Explain the importance of reduced NADP in the process of photosynthesis. [2]

1. **Reduced NADP provides reducing power;** for the calvin cycle of photosynthesis/dark reaction to take place;
2. **converts glycerate-3-phosphate to glyceraldehyde-3-phosphate OR supplies hydrogen to reduce GP to TP** and regenerates RuBP

The length of root hair in wild and mutant, *A. thaliana* plants were measured at intervals, together with the activity of reduced NADP oxidase in the tips of the root hairs. The results are shown in Fig 2.3

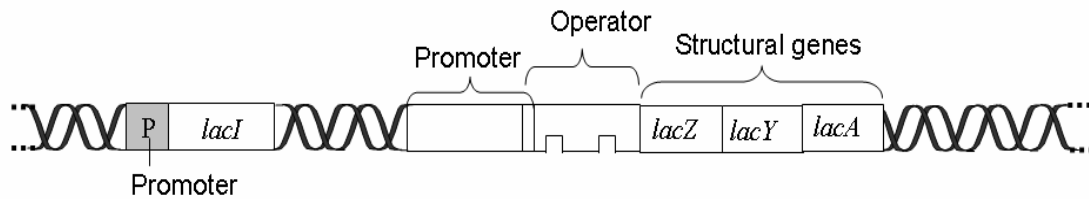


(c) With reference to figure 2.3 above, suggest an explanation, making use of your knowledge and understanding of photosynthesis, for the difference in the growth of root hair in the two types of plant. [5]

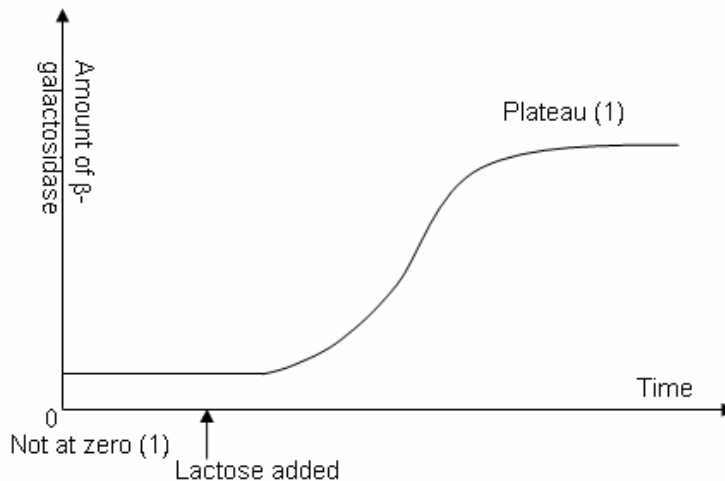
1. **Both types of root hairs grow at same rate for the first 40 minutes; due to similar mean activities of NADP oxidase is the similar (wild type = 12 au, mutant = 8au)**
2. Following this **growth rate of wild type continues to increase (12 to 38um)** matched in comparison to the **Mean activity of NADP oxidase** for the wild type increases from **13 -42 au**
3. while the **mutant type levels off (plateaus off at 12um)** with enzyme activity decreasing from 9 to 3/4 a.u)
4. **Reduced NADP oxidase converts reduced NADP to NADP** which is required for the light reactions in photosynthesis;
5. **Produces more energy in form of ATP/Increased ATP formation in the light reaction; ATP is required for cellular growth/ increase in cell number etc; /For the growth of the root hairs which is energy requiring.**

[Total: 13]

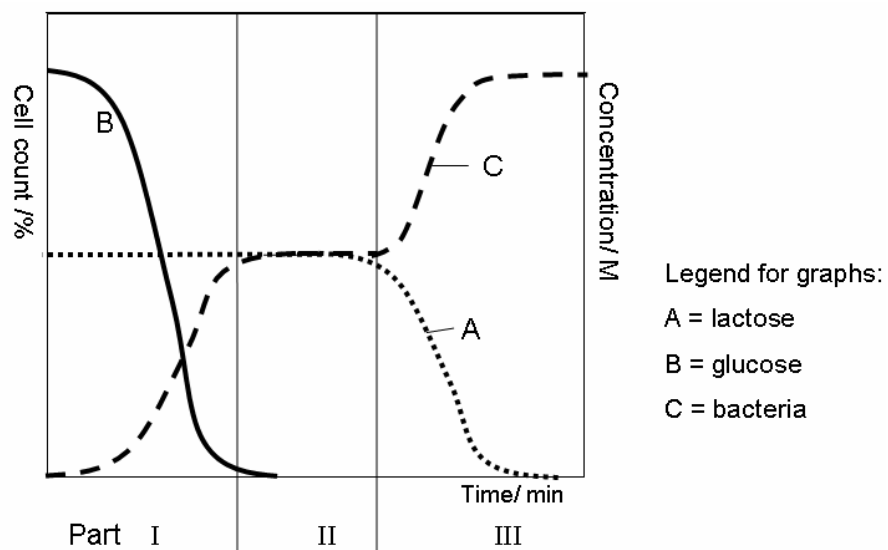
- 3 A study of the bacterium's chromosome showed that it possessed a *lac* operon to allow the bacterium to metabolise lactose. The structure of the *lac* operon is shown below:



- (a) On the axes below, draw a graph showing how the amount of β -galactosidase in the bacterium is affected by the addition of lactose. [2]



- (b) The graph below shows the growth of bacteria in the presence of glucose and lactose.



Describe and explain the three parts of graph C from the view point of the Lac Operon.

Part I:

1. The Lac Operon controls the transcription of Lac structural genes which see to the utilization and breakdown of Lactose. With lactose present, **allolactose** (Isomer of lactose) binds to the repressor protein **changing its configuration**.
2. With the change in configuration the **repressor protein is now unable to bind to the operator region** of the promoter of the Lac Operon.
3. The presence of **Glucose** in solution causes the **levels of cyclic AMP to be extremely low**,
4. as **glucose causes the efflux of cAMP** and **hinderance of synthesis of cAMP**.

5. cAMP when combined with cAMP Receptors become **activated CRP** which act as **transcription factors to increase the level of transcription when associated with RNA polymerase.**
6. With the repressor protein not present, **RNA polymerase is free to bind to the promoter** but due to **low activated CRP levels and transcription of lac genes can take place but at very low rates of transcriptions.**
7. **Glucose thus is used as the main respiratory substrate.**
8. **cell count increases steeply** as glucose is available / used as substrate

Part II:

[3]

1. **cell count remains constant** after glucose has been used up
2. **time is needed for the expression/activation of the lac operon** to produce **β -galactosidase** before lactose can be used/metabolized
3. With **Glucose depleted**, levels of **cAMP rise**, causing also the levels of **activated CRP to rise** as well **allowing for greater rates of transcription** of Lac structural genes by RNA polymerase.

Part III:

[5]

1. As **β -galactosidase** is produced more and more lactose is broken down to galactose and glucose and used as respiratory substrates- cell count increases again as lactose/broken down and used in metabolism
2. as more lactose is used up, **less allolactose is available to bind to the repressor protein.**
3. as a result, the **repressor protein is now free to bind to the operator** of the promoter.
4. This **hinders binding of RNA Polymerase to the promoter** and **transcription stops**, and **β -galactosidase** production stops as well.
5. **graph plateaus** as lactose is used up

[Total:16]

- 4 A series of breeding experiments were carried out with *Drosophila*, the fruit fly. A cross between pure breeding males with red eyes and straight wings and females with purple eyes and curled wings produced offspring all with red eyes and straight wings. The same results are obtained when the parents of the same pure-breeding phenotypes were used again but with the sexes reversed.

- (a) Suggest a reason for carrying out the cross with the sexes reversed. [1]
1. To check whether the genes are sex-linked

Female offspring from the first experiment were then crossed with purple eyed curled winged males.

- (b) Draw a genetic diagram to show the results of this cross. [4]

Let allele for red eye be R
Let allele for purple eye be r

Let allele for straight wing be S
Let allele for curled wing be s

Parental genotype: Female RrSs Male rrss

Meiosis

Parental Gametes: RS, Rs, rS, rs rs

Offspring genotype: RrSs, Rrss, rrSs, rrrs

Genotypic ratio: 1 : 1 : 1 : 1

Phenotype: Red eye straight wing, red eye curled wing, purple eye straight wing, purple eye curled wing

Phenotypic ratio: 1:1:1:1

The progeny listed below were produced from this cross.

173 red eye, straight wing
165 red eye, curled wing
169 purple eye straight wing
179 purple eye, curled wing

1 Mark

1 Mark

1 Mark

1 Mark

A null hypothesis was put forward that the genes for eye colour and wing shape are not linked. The chi-squared (χ^2) test was applied to the data obtained. The formula is given below:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- (c) (i) Complete the Table and use the formula to determine the value of χ^2 for the results of this investigation. Show working. [2]

Phenotype	red eye, straight wing	red eye, curled wing	purple eye straight wing	purple eye, curled wing
Observed	173	165	169	179
Expected	171.5	171.5	171.5	171.5

1 mark for a full correct table

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$= [(171.5-173)/171.5 + (165-171.5) / 171.5 + (169-171.5)/171.5 + (179-171.5)/171.5]$$

$$= 0.01 + 0.24 + 0.04 + 0.33$$

$$= 0.62$$

Answer: 0.62

1 mark for correct calculation and answer

- (ii) Use the table to determine if this hypothesis was correct. Explain how you arrive at your answer. [3]

Degrees of freedom	Probability value					
	0.99	0.95	0.10	0.05	0.01	0.001
1	0.0002	0.0039	2.71	3.84	6.63	10.83
2	0.020	0.103	4.61	5.99	9.21	13.82
3	0.115	0.352	6.25	7.81	11.34	16.27
4	0.297	0.711	7.78	9.49	13.28	18.47

$$\chi^2_{cal} < \chi^2_{tabulated},$$

Degree of Freedom: No of Characteristics -1
 $\nu = 4 - 1 = 3$

Probability : $P = 0.05$

$\chi^2_{tabulated}$ From the Table: **7.81**

$$0.62 < 7.81$$

Since $\chi^2_{cal} < \chi^2_{tabulated}$,

Hence difference is not significant and is due to chance

Accept null hypothesis i.e. there is no difference between the observed and expected numbers

1 Mark

1 Mark

[Total : 10]

- 5 In most cases of chronic myeloid leukemia, the cells share a chromosome abnormality not found in other non-leukemic white blood cells, or any other cells in the body. This abnormality is shown in Fig. 5.1 below.

The Philadelphia chromosome (Ph) is an abnormally short chromosome 22 that is one of the two chromosomes involved in genetic exchange with chromosome 9. It is the first consistent abnormality found in any kind of malignancy.

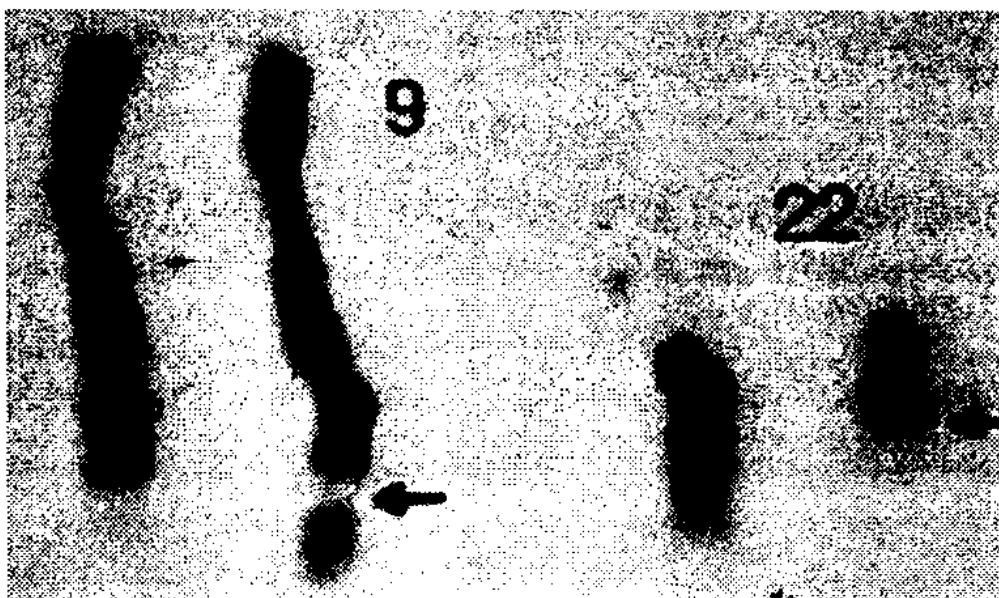


Fig. 5.1

- (a) Deduce the kind of genetic exchange that occurs between chromosome 9 and 22 in 5.1. [1]
1. Translocation

The DNA removed from chromosome 9 contains most of a gene titled **c-ABL**. c-ABL produces an ABL protein that acts as a tyrosine kinase. The break in chromosome 22 occurs in the middle of a gene titled **BCR**. The resultant Ph chromosome has the BCR fused to most of c-ABL.

Transcription and translation of the hybrid BCR-ABL gene produces an abnormal 'fusion' protein that activates **constitutively** (i.e. all the time) a number of cell activities that are normally turned on only in the presence of a growth factor, such as a Platelet Derived Growth Factor (PDGF). The fusion protein also functions as a tyrosine kinase. This mutation occurs in a single bone marrow cell, and by mitosis, the number of Ph chromosome containing cells increases.

- (b) (i) Based on the description of the fusion protein, what can one deduce about the type of gene the c-ABL belongs to? Explain your answer. [1]
1. **Proto-oncogene; codes for protein involved in cell signaling; gains function on mutation.**
- (ii) Predict two effects this hybrid protein has on the cell. [1]
1. **increase rate of mitosis/ cell growth/ uncontrolled cell growth.**
 2. **cell escapes cell death/apoptosis.**
 3. **cells do not show contact inhibition;**
 4. **Cell grows independent of growth factors;**

During the chronic phase of this disease, these cells are still able to exit the cell cycle and perform their normal function.

- (c) Using your knowledge of cell signaling, describe the sequence of events that occurs to a cell for it to become differentiated. [3]

Reception

1. Cues can be **hormones, growth factors, ligand** etc; attach to **receptors on cell surface membrane;**

Transduction

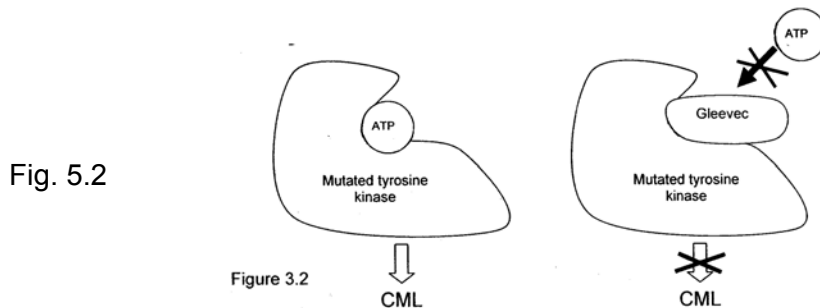
2. Information is transduced into cell through **conformational changes_of the receptor protein;**
3. **Phosphorylation** cascade through **kinases;** protein kinase (move into nucleus to) **activate transcription factors;**

Cellular response

4. Switch genes on/**activates genes/ stimulates transcription; Specific proteins/ tissue-specific structures are produced;**

At some point another mutation in a tumour suppressor gene or a proto-oncogene will occur in one of these cells. This additional mutation causes the rate of mitosis in these cells to increase sharply. The daughter cells fail to differentiate and a patient enters the crisis phase of the disease.

Presently, a drug named Gleevec[®] is being used to treat patients with CML. Patients given Gleevec[®] have so far shown return to normal blood counts and disappearance of leukemia cells in the marrow. Fig. 5.2 shows how Gleevec works.



- (d) Describe the role of tumour suppressor genes in normal cells. [2]
1. Tumour suppressor genes are involved in **DNA repair/fixing damaged or broken DNA**
 2. Tumour suppressor genes cause **mutated cells to undergo apoptosis**.
- (e) Predict whether a mutation in the tumour suppressor gene will be a dominant or recessive mutation. Give reasons for your answer. [2]
1. **Recessive mutation.**
 2. **Loss of function mutation**, which means that the **normal, non-mutated gene can still function → synthesise tumour suppressor protein.**
 3. **Both genes must be mutated for tumour suppressor protein to be non-functional.**
- (f) Name one other way in which a cell's ability to proliferate can become uncontrolled. [1]
1. **Re-activation of telomerase**
- (g) Briefly explain the mode of action of Gleevec in controlling CML. [2]
1. **Competitive inhibitor**
 2. **Competes with ATP for active site of mutated tyrosine kinase, prevents binding of ATP.**
 3. **Prevents phosphorylation cascade/cell signaling processes.**

Recently two different point mutations have been detected in the ATP binding site which affects the binding of Gleevec, leading to resistance. Table 5.3 summarises the two types of mutations that have been detected.

Table 5.3 Showing the types of mutations to ABL-BCR and their effects

Types	Mutation 1	Mutation 2
Change of base	thymine to cytosine	cytosine to thymine
Amino acid change	methionine to threonine	threonine to isoleucine
Resistance to Gleevec	partial	complete
Recommended treatment	higher dosage of Gleevec	another drug

- (h) How does a single base change lead to resistance to Gleevec? [2]
1. **change of sequence of bases (in gene) causes change in codon;**
 2. **different amino acid coded for resulting in primary structure of polypeptide change/ change amino acid sequence;**
 3. **different conformation or shape of active site/ ATP binding site;**
 4. **not complementary to Gleevec/ Gleevec cannot bind to ATP-binding site;**
- (i) Suggest possible reasons for the differences in the resistance to Gleevec and the recommended treatment despite both mutations involving one amino acid change. [2]
1. **different amino acids have different R groups/ properties/structure;**

2. mutation 1 – **properties between the 2 amino acids may be more alike** so Gleevec can still fit but not ideally;
3. mutation 2 – **properties between the 2 amino acids very different, isoleucine may hinder binding of Gleevec because of steric hindrance/ repel Gleevec**

[Total: 17]

- 6 (a) Two closely related species of frog, *Hyla ewingi* and *Hyla verrauxi* live in South Australia.

Explain what is meant by the terms;

(i) *Niche*

[2]

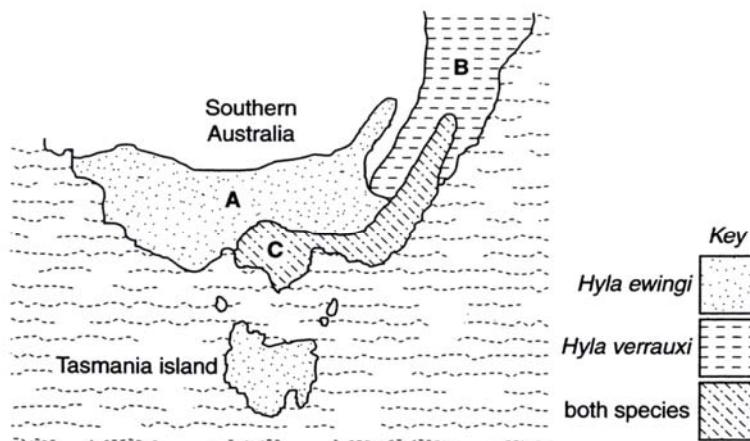
1. Niche is the **unique position of an organism in an ecosystem**
2. It include **descriptions of the organism's life history, habitat and food chain**
3. It describes **how a population responds to the abundance of its resources and enemies.**
4. Niche also describes **how the various abiotic factors influences how populations affect, and are affected by resources and enemies**
5. **No 2 species can occupy in the same niche for a long time.**

(ii) *Species*

[3]

1. Similar **morphological, physiological, biochemical** and behavioural features;
2. **DNA** of organisms of a **species nearly identical/have identical gene loci.**
3. Interbreed/ **reproduce to produce fertile offspring;**
4. **Reproductively isolated;**
5. Occupy **same (ecological) niche;**

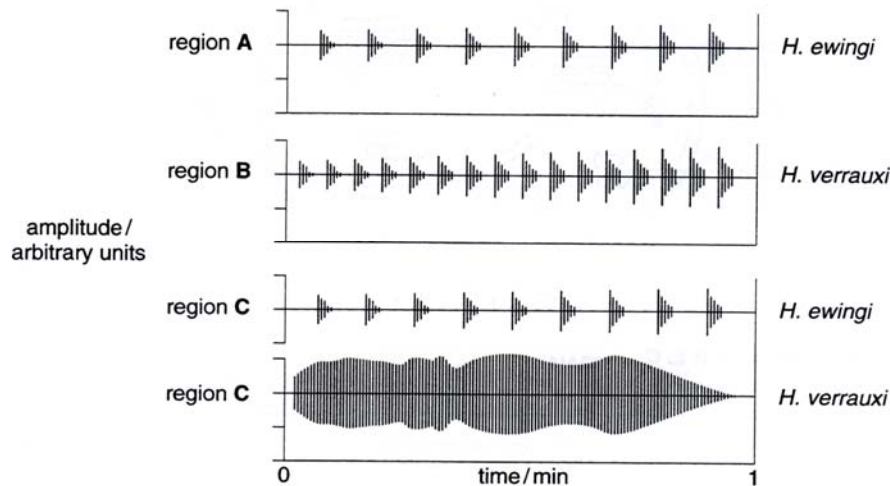
The figure below shows the distribution of the tree frogs in Southern Australia.



Hyla ewingi and *Hyla verrauxi* are two closely related species of tree frogs from southern Australia. Research from breeding studies and DNA sequence data has shown that they have weak genetic incompatibility.

Male frogs attract females of the same species for mating by their pulsing call. The pulse rate of the male calls of the two species is almost identical. However, when both species coexist within the same region, the calls of *H. ewingi* are quite different than those of *H. verrauxi*.

The following data shows the calls of *H. ewingi* and *H. verrauxi* from three regions, A, B and C. The male advertisement calls were recorded and played back electronically to produce the pattern shown



Some female frogs of the species *H. verrauxi* were transferred from region B to region C. Observations shown that they were not attracted by the calls of the males in region C.

(b) Suggest why this is so. [3]

1. The male advertisement calls by *H. verrauxi* in region C are no longer pulsatile / discontinuous
2. More continuous without any interval in C;
3. Amplitude of the sound waves also differ: Sound intensity of male call in region B is from soft to loud compared to loud to soft in C.

(c) Explain how the process of natural selection could have led to the different male advertisement call in *H. verrauxi* in region C. [5]

- 1 Variation in male advertisement calls present in population of *Hyla* frogs in region C;
- 2 As *H. ewingi* and *H. verrauxi* have similar advertisement calls; competition for mates exists between these two species;
- 3 Competition for mates (sexual selection by females) in the same habitat selects males with a different/ continuous calls compared to males with a pulsatile call;
- 4 Successful male mates with female *Hyla verrauxi* tree frogs;
- 5 reproduced more offspring with similar male advertisement call;
- 6 As these genes that code for such male advertisement calls are passed down to the next generation;
- 7 Those *H. verrauxi* males who fail to mate with another female will diminish in numbers after several generations;
- 8 Over time, region C sees an evolution of a different male advertisement call for *H. verrauxi*

[Total: 13]

Section B

Answer **ONE** question **ONLY**.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 7 (a) Describe and explain the secondary structure of proteins? [6]
- 1 derived from the **primary sequence of aa**
 - 2 Secondary structures are either of **α helices**; **β -pleated sheets**; Shape is maintained by **weak H-bonds**.
 - 3 H-bonds are intra molecular /intra chain for **α -helices between every forth amino acid/aa 1 with aa 5**; comprises
 - 4 a weakly **positive H atom in the $-NH$ gp** and a **weakly negative oxygen atom in the $-CO$** ;
 - 5 **β -pleated sheets** can be arranged in a parallel or anti-parallel orientation;
 - 6 comprise of weak H bonding between **positive H atom in the $-NH$ gp** and a **weakly negative oxygen atom in the $-CO$**
- (b) Describe the main ways in which a globular protein differs from DNA. [8]

Pt of comparison	Globular proteins	DNA
1 Shape	Compact & round globular	Linear
2 Solubility	Yes	No/less soluble
3 Monomers	Amino acids	Nucleotides
4 Types of monomers	20 used in humans	4 types (A, T, G, C)
5 Bonding 1	Linked by peptide bonds	Linked by phosphodiester bonds
6 Number of subunits	One or more	Always 2 under physiological conditions
7 Bonding 2	Polypeptide chains/subunits linked by ionic bonds, covalent disulphide bonds, hydrophobic interactions and hydrogen bonding	Only hydrogen bonding links the two DNA chains
8 Function	Multiple functions: Enzymatic, structural, transport, energy source (any 2)	Carry hereditary information/code for synthesizing products
9 What confers functionality	Its function is usually linked to its 3D shape/ conformation and configuration	Its function is dependent only on the order in which the monomers are arranged in DNA chain
10 Levels of organization	4 levels	1 structure/ α –helix
11 Charge	Can be of any charge	Negatively charged
12 Location	Everywhere/both inside and outside the cell	Only inside the cell; R DNA only in nucleus (found in cytoplasm)

- (c) Using HIV protease as an example, analyze how its structure is related to its function. [6]
1. a small **globular enzyme** found in HIV – **soluble** in water
 2. consists of **2 identical polypeptide chains**, each only 99 amino acids long
 3. each monomer contains an extended **β -sheet region known as a flap**.
 4. **active site is located where the two monomers meet** and is "covered" by the flaps that extend
 5. the specific task of the flaps is **guarding the entrance to the active site and excluding incoming water** molecules, which may interfere with proteolytic cleavage. (Essential for catalytic activity is a water molecule already bound to the backbone).
 6. **flaps are flexible and allow a substrate access** to the active site by opening and folding
 7. high level of **complementarity** is attained by way of protein changing its conformation to accommodate the substrate.

[Total:20]

OR

- 8 (a) Describe the structure and function of the parts of a eukaryotic chromosome. You may wish to include a diagram as part of your description. [10]

Component	Structure: 1 mark / component	Function: 1 mark / component
Chromatids	2 sister chromatids / a pair of chromatids / identical DNA sequences / genetic information on a pair of sister chromatids	Distribution of one copy to each new daughter cell
Centromere	Noncoding / uncoiled / narrow / constricted region / determines arm ratio of chromosomes	Joins / holds / attaches sister chromatids together
Nucleosome concept	Composed of eight histone molecules united to form a core / spool. 146bp of DNA wound around the histone core. Between nucleosomes: variable amount of non-nucleosomal "linker" DNA	Packaging / compacting Exposed to nuclear environment, ∴ accessible to proteins involved in its duplication and regulation of its expression
Chromatin form (heterochromatin/euchromatin)	Condensed / supercoiled (heterochromatin) Loosely coiled (euchromatin)	Proper distribution in cell division (NOT during replication); facilitates packing of DNA in nucleus Gene expression during interphase / replication occurs when DNA is loosely packed
Kinetochores	Disc-shaped proteins situated in the centromere of a chromosome	Attachment / alignment of chromatids to spindle fibres during nuclear division
Genes or DNA		Codes for proteins
Telomeres	Tips / ends of a chromosome; Consist of noncoding repetitive sequences	Protection against degradation / aging; Limits number of cell division – Hayflick limit

- (b) Explain how prokaryotic genome structure differs from eukaryotes. [6]

Genome Level:

1. Prokaryotic genome is **circular** while eukaryotic **genome is linear**.
2. Prokaryotic genes are **grouped into operons** with one promoter controlling **expression of many genes** while eukaryotic genes **are not grouped into operons**, each **gene has its own promoter**.
3. Prokaryotic DNA is **free floating in the cytoplasm** as a nucleoid. Eukaryotic genome exists in the **nucleus as distinct chromosomes**.
4. Prokaryotic DNA is made up mostly of **coding regions**, with **few repetitive/satellite DNA**. Only **3% of eukaryotic DNA is coding**. Most of it is **satellite DNA/repetitive DNA**.
5. Prokaryotic DNA is **not coiled** round **histone proteins**, only **supercoiled**. **Eukaryotic DNA is coiled around histone proteins and further folded and coiled around scaffold proteins**.
6. Prokaryotic DNA has **no telomeres**, while eukaryotic DNA **has telomeres at the end** to prevent DNA degradation due to replication.

(c) Discuss the evolutionary significance of organizing genes into chromosomes.

[4]

1. **Eukaryotes have genes organized to chromosomes** because they are **more complex, require better regulation of gene expression**.
2. **They also have larger genomes hence a need to replicate them fast provides evolutionary advantages.**
3. **Specialisation – Each chromosome** has genes coding for similar **traits/set of traits. Allows faster expression of a group of traits.**
4. **Presence of silencer genes and activator genes on different chromosomes** allows **better control of transcription**. (Activator proteins bind to enhancers while silencer proteins bind to silencers)
5. **Faster replication** as different chromosomes can be **replicated simultaneously**.
6. **More efficient proof reading**, as many **DNA polymerase I** can **proofread different chromosomes** at the same time.